

Derek Wu

Senior Backend Engineer

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Profile

Senior Backend Engineer with over 5 years of experience in backend systems, infrastructure engineering, and practical machine learning deployment within startup environments. Experienced in distributed systems, cloud infrastructure, and open-source software development, with strong proficiency in Python, Rust, and cloud-native technologies.

- **Work Authorization:** Currently based in Taiwan and authorized to work in Japan through Dec. 2026 under a valid Working Holiday Visa.
- **Infrastructure Experience:** Led infrastructure migration from Kubernetes to Docker-based virtual machines, reducing cloud infrastructure costs by 53%.
- **Open Source & Community:** Maintainer of open-source audio and computer vision projects with over 2.6k combined GitHub stars. Active contributor to the Python community through PyCon TW and PyCast.
- **Objective:** Seeking backend, infrastructure, or platform engineering opportunities in Japan, particularly within international and engineering-driven teams.

Highlighted Projects

- omnizart** 📖 **Creator & Core Maintainer** Python, TensorFlow, Signal Processing
 - Built a unified, production-ready audio intelligence toolbox showcasing near **2,000 GitHub stars**; translated complex laboratory code into clean, modular library architectures consumed worldwide.
- oemer** 📖 **Creator & Core Maintainer** Python, Computer Vision, Semantic Segmentation
 - Architected and open-sourced one of the earliest end-to-end Optical Music Recognition system (**730+ GitHub stars**), successfully engineering complex digital signal processing pipelines for computer vision model inference.

Experience

- 03/2025 – Present 📖 **Independent Software Engineer & Open Source Contributor, Taiwan & Japan**
 - Relocated independently to Japan under a Working Holiday Visa, developing practical Japanese communication skills through daily life and work experience. Obtain JLPT N3 qualification 07/2025.
 - Worked in a fast-paced team environment at GALA ski resort (12/2025 – 03/2026), adapting to Japanese workplace communication and cross-cultural collaboration.
 - Active volunteer within the Python community through PyCon TW and PyCast, including co-hosting technical podcast content and interviewing open-source developers at PyCon 2025.
 - Developed several open-source Rust utilities (thsr-ticket-rs, bfx-rs, BigBench, calc-pi) to deepen systems programming and concurrent programming experience.

Experience (continued)

- 12/2022 – 02/2025  **Backend & AI Engineer, *Transferhelper***
- Led infrastructure migration from Kubernetes to Docker-based virtual machines, reducing cloud infrastructure costs by 53%.
 - Architected and executed algorithmic lending strategies on Bitfinex, consistently yielding a 15% APR over a six-month period.
 - Engineered backend microservices utilizing PostgreSQL, Redis, and Kubernetes on Google Cloud Platform (GCP).
 - Established system observability pipelines through Prometheus, Grafana, and Slack alert integrations.
 - Automated CI/CD pipelines via Google Cloud Build, achieving unit test coverage exceeding 90%.
 - Researched and implemented deep learning pipelines for reconstructing 3D models from 2D human photographs.
- 01/2022 – 08/2022  **Backend Engineer, *Pinkoi***
- Led the architectural refactor and major upgrade of a legacy coupon logic system, effectively reducing code complexity and improving system performance.
 - Partnered with the cross-functional finance team to design and deploy a new automated invoice billboard feature.
 - Demonstrated strong tracing capabilities to rapidly navigate complex codebases, isolating critical production bugs and delivering precise hotfixes under tight timelines.
- 07/2021 – 11/2021  **Backend Engineer, *Meteo Piano***
- Proposed, developed, and open-sourced the earliest end-to-end Optical Music Recognition (OMR) solution, frameworked as “*oemer*” on GitHub.
 - Integrated disparate machine learning components into a highly available, distributed RESTful API server within a strict one-month deadline.
 - Deployed and managed scalable services on AWS cloud infra, utilizing EC2, S3, VPC, ECS, and Application Load Balancers.
- 06/2017 – 12/2020  **Research Assistant, *IIS Academia Sinica***
- Advanced state-of-the-art research in note-level multi-instrument music transcription, publishing two IEEE conference papers and authoring one IEEE journal paper.
 - Consolidated isolated laboratory research models into a unified, production-ready Python package, *omnizart*, which garnered near 2,000 GitHub stars.
- 07/2019 – 06/2020  **Backend Engineer Intern, *TrendMicro***
- Resolved a long-standing, complex cross-team bottleneck within two months by securely bridging internal ADFS authentication with AWS IAM services.
 - Optimized deployment pipelines on AWS, integrating a robust Blue/Green deployment strategy for zero-downtime releases.
 - Streamlined legacy deployment scripts for readability, implementation of unit tests, and reduced overall CI/CD runtime by up to 50%.
 - Ported high-performance machine learning execution blocks from Python to Java to ensure seamless backend infrastructure integration.

Education

- 2018 – 2020  **M.Sc. Computer Science**, National Taiwan Normal University.
Thesis title: *Multi-Instrument Automatic Music Transcription With Self-Attention-Based Instance Segmentation*.
Publications: Published 3 IEEE papers (including 1 IEEE Journal paper) on deep-learning-based music transcription models.
- 2014 – 2018  **B.Sc. Computer Science**, National Taiwan Normal University.

Skills

- Languages  Native Mandarin Chinese speaker; strong proficiency in reading and writing English.
- Coding  Python, FastAPI, Rust, asynchronous programming.
- Databases  PostgreSQL, MySQL.
- Infra  Google Cloud Platform (GCP), Amazon Web Services (AWS), Docker, Kubernetes, GitHub, Redis, Nginx, Linux.
- Misc.  Academic research, machine learning, TensorFlow, PyTorch, scikit-learn, NumPy, OpenCV.